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Layered Heterostructure Ionogel Electrolytes for High-Performance Solid-State Lithium-Ion Batteries

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Ionogel electrolytes based on ionic liquids and gelling matrices offer several advantages for solid-state lithium-ion batteries, including nonflammability, wide processing compatibility, and favorable electrochemical and thermal properties. However, the absence of ionic liquids that are concurrently stable at low and high potentials constrains the electrochemical windows of ionogel electrolytes and thus their high-energy-density applications. Here, ionogel electrolytes with a layered heterostructure are introduced, combining high-potential (anodic stability: >5 V vs Li/Li^+) and low-potential (cathodic stability: <0 V vs Li/Li^+) imidazolium ionic liquids in a hexagonal boron nitride nanoplatelet matrix. These layered heterostructure ionogel electrolytes lead to extended electrochemical windows, while preserving high ionic conductivity (>1 mS cm^{-1} at room temperature). Using the layered heterostructure ionogel electrolytes, full-cell solid-state lithium-ion batteries with a nickel manganese cobalt oxide cathode and a graphite anode are demonstrated, exhibiting voltages that are unachievable with either the high-potential or low-potential ionic liquid alone. Compared to ionogel electrolytes based on mixed ionic liquids, the layered heterostructure ionogel electrolytes enable higher stability operation of full-cell lithium-ion batteries, resulting in significantly enhanced cycling performance.

1. Introduction

Lithium-ion batteries are the leading energy storage technology for the growing markets of portable electronics, electric vehicles, and grid-level management systems.[1,2] To satisfy the increasing demand for higher energy densities, substantial effort has been devoted to elevating the operating voltage of lithium-ion batteries. Although a number of high-potential cathode materials have been developed in this context, their practical deployment has been hindered by insufficient high-potential stability of conventional liquid electrolytes based on carbonate solvents and lithium salts. Moreover, the high flammability of organic solvents poses serious safety concerns when liquid electrolytes are subjected to voltage conditions exceeding their electrochemical stability windows, which presents a further barrier to increasing energy density in conventional lithium-ion battery designs. To overcome these issues, significant attention has been directed toward the development of solid-state electrolytes as a replacement to liquid electrolytes in lithium-ion batteries.[3,4] Although considerable progress has been achieved, solid-state electrolytes based on inorganics and polymers continue to face important challenges in practical applications, including low ionic conductivity, high interfacial resistance, and cumbersome processing.

Ionogels are solid-state electrolytes based on ionic liquids and gelling matrices, which have attracted considerable attention for lithium-ion batteries.[5,6] In contrast to traditional liquid electrolytes, ionic liquids offer nonflammability, negligible vapor pressure, and high thermal stability, which not only address safety concerns but also elevate the high-temperature limit of battery operation.[7,8] Moreover, ionogel electrolytes provide high ionic conductivity, favorable interfacial contact with electrodes, and wide processing compatibility, which address the key issues confronting inorganic and polymer solid-state electrolytes.[9–12] The electrochemical stability windows of ionogel electrolytes primarily depend on the ionic liquids. Since the anodic and cathodic stability of ionic liquids can be manipulated by altering the constituent anions and cations, a wide range of ionic liquids with different electrochemical windows have been explored for lithium-ion batteries.[13,14] However, despite extensive research into ionic liquid electrolytes, no single ionic liquid has simultaneously achieved desirable high-potential and low-potential stability. Consequently, full-cell lithium-ion batteries based on ionic liquids have typically been demonstrated using electrodes with modest potentials,[15–17] thereby restricting operating voltage and energy density. While mixed ionic liquids have been proposed for synergetic effects, this approach has not fully widened the operating voltage of

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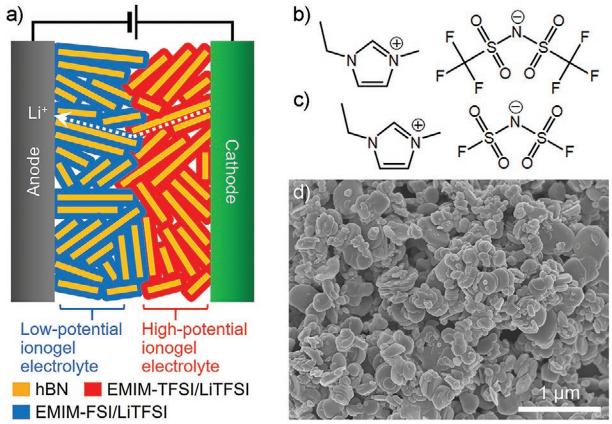

Figure 1. Layered heterostructure ionogel electrolyte for solid-state lithium-ion batteries. a) Schematic of a solid-state lithium-ion battery using the layered heterostructure ionogel electrolyte with two different ionic liquids and hexagonal boron nitride (hBN) nanoplatelets. EMIM-TFSI, EMIM-FSI, and LiTFSI denote 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide, 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)imide, and lithium bis(trifluoromethylsulfonyl)imide, respectively. In the layered heterostructure ionogel electrolyte, the ionogels based on EMIM-FSI/LiTFSI and EMIM-TFSI/LiTFSI serve as low-potential and high-potential ionogel electrolytes, respectively. b,c) Chemical structures of EMIM-TFSI (b) and EMIM-FSI (c) ionic liquids. d) Scanning electron microscopy image of the hBN nanoplatelets employed as the solid matrix for the formulation of the ionogel electrolytes.

full-cell lithium-ion batteries.[18–20] Alternatively, organic carbonates have been added to ionic liquid electrolytes to enhance cathodic stability, but at the cost of compromising safety.[21–23]

Here, we report the development of layered heterostructure ionogel electrolytes based on two imidazolium ionic liquids to broaden the electrochemical stability window and thus operating voltage of full-cell solid-state lithium-ion batteries. The key to realizing layered heterostructures is to increase ionogel mechanical properties in a manner that minimizes intermixing at the layered ionogel heterointerface through the use of hexagonal boron nitride (hBN) nanoplatelets as the gelling matrix. In the optimal design for full-cell solid-state lithium-ion batteries, one of the ionic liquids possesses high anodic stability and adjoins the cathode, while the ionic liquid on the other side of the ionogel heterointerface possesses high cathodic stability and contacts the anode. In this manner, layered heterostructure ionogel electrolytes enable extended electrochemical windows, while maintaining high ionic conductivity, which result in improved operating voltages and rate performance of solid-state lithium-ion batteries.

2. Results and Discussion

Figure 1a depicts a schematic of a lithium-ion battery using the layered heterostructure ionogel electrolyte based on two ionic liquids and hBN nanoplatelets. To fabricate the layered heterostructure ionogel electrolyte, one ionogel was prepared with a 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide (EMIM-TFSI, Figure 1b) ionic liquid containing 1 м lithium bis(trifluoromethylsulfonyl)imide (LiTFSI) salt, which serves as a high-potential ionogel electrolyte on the cathode side of the battery. The other ionogel was produced with a 1-ethyl-3-methylimidazolium bis(fluorosulfonyl)imide (EMIM-FSI, Figure 1c) ionic liquid containing 1 м LiTFSI salt, which serves as a low-potential ionogel electrolyte on the anode side of the battery. Imidazolium-based ionic liquids were selected since their low viscosity is favorable for fast lithium-ion transport.[24] Furthermore, EMIM-TFSI was used for the high-potential ionogel electrolyte due to the oxidative stability of TFSI anions[15] and EMIM-FSI was employed for the low-potential ionogel electrolyte because FSI anions enable cathodic stability.[25,26] Although the different ionic liquids were employed for the high-potential and low-potential ionogel electrolytes, the same lithium salt with the identical concentration was used to induce a uniform distribution of lithium ions across the layered heterostructure ionogel electrolytes and thus facilitate lithium-ion transport.

Liquid-phase exfoliated hBN nanoplatelets (Figure 1d) were used as the gelling solid matrix since hBN is electrically insulating, chemically inert, thermal stabile, and mechanically robust.[27] In addition to these intrinsic material benefits, exfoliated hBN nanoplatelets provide large surface area that strongly confines the ionic liquids, which is confirmed by the negligible fluidic behavior (Figure S1, Supporting Information) of the ionic liquids in the ionogels. Consequently, both the high-potential and low-potential ionogel electrolytes showed high mechanical strength (storage modulus >1 MPa, Figure S2, Supporting Information), eliminating the need for a separator in the battery assembly. Moreover, the hBN nanoplatelets lead to not only effective physical confinement but also favorable ion conduction,[27] allowing the construction of the layered heterostructure ionogels with high room temperature ionic conductivity (>1 mS cm^{-1}, Figure S3, Supporting Information).

Cyclic voltammetry (CV) was performed with Li|ionogel electrolyte|stainless-steel cells to investigate the electrochemical stability windows of the high-potential and low-potential ionogel electrolytes. **Figure 2**a displays CV curves for the high-potential and low-potential ionogel electrolytes, which were acquired between 3.0 and 5.0 V (vs Li/Li^+) to observe their anodic stability. The high-potential ionogel electrolyte exhibited stable CV curves with negligible change in the current density, revealing anodic stability over 5 V (vs Li/Li^+, Figure S4, Supporting Information). In contrast, the low-potential ionogel electrolyte showed significant decomposition with increasing current density at >4 V (vs Li/Li^+), originating from the oxidation of FSI anions.[20] To further confirm the anodic stability, half-cells with lithium nickel manganese cobalt oxide ($LiNi_{0.33}Mn_{0.33}Co_{0.33}O_2$, NMC) were tested using the high-potential and low-potential ionogel electrolytes, with a voltage range of 2.5–4.3 V (vs Li/Li^+) at 0.1C. As shown in Figure 2b, the charge cutoff voltage is lower than the anodic limit of the high-potential ionogel electrolyte, but exceeds the anodic limit of the low-potential ionogel electrolyte, resulting in electrolyte oxidation. Accordingly, the NMC half-cell (Figure 2c) with the high-potential ionogel electrolyte showed stable operation with typical voltage profiles and an initial discharge capacity of 146 mAh g^{-1}, whereas the NMC half-cell (Figure 2d) with the low-potential ionogel electrolyte experienced severe degradation at >4 V (vs Li/Li^+) and did not reach the charge cutoff voltage, in agreement with the CV characterization.

Figure 3a compares CV curves of Li|ionogel electrolyte|stainless-steel cells with the high-potential and

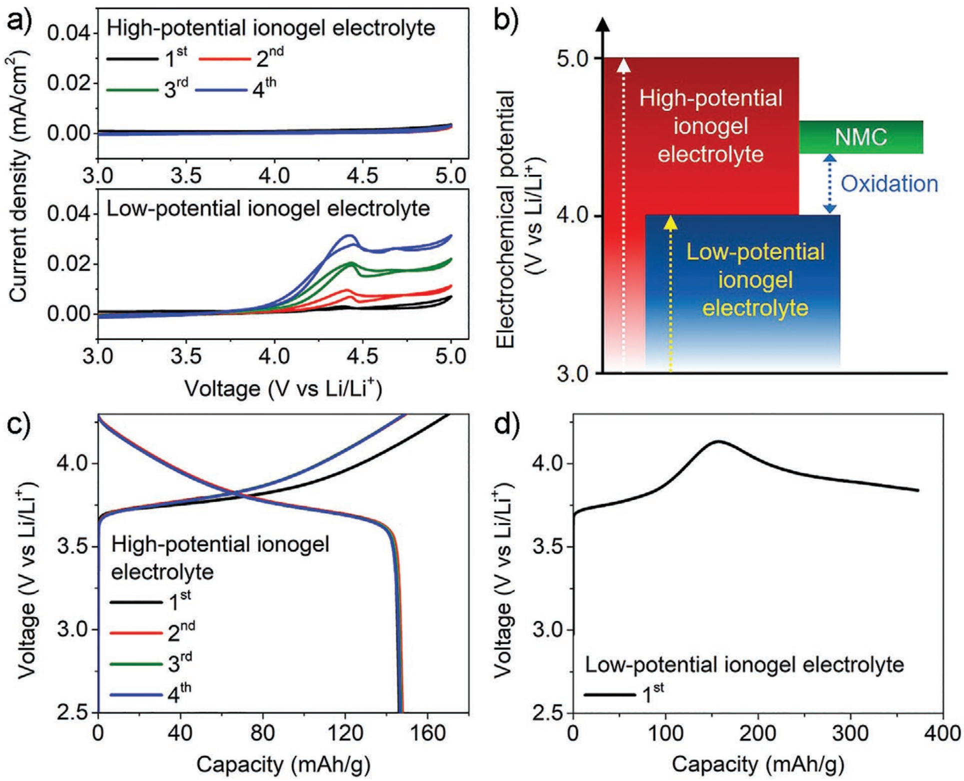

Figure 2. Anodic stability of the high-potential and low-potential ionogel electrolytes. a) Cyclic voltammograms of Li|ionogel electrolyte|stainless-steel cells with (top) the high-potential and (bottom) low-potential ionogel electrolytes between 3.0 and 5.0 V (vs Li/Li^+) at a scan rate of 1 mV s^{-1}. b) Schematic diagram of the anodic stability of the high-potential and low-potential ionogel electrolytes for lithium nickel manganese cobalt oxide ($LiNi_{0.33}Mn_{0.33}Co_{0.33}O_2$, NMC) half-cells. c) Charge–discharge voltage profiles of an NMC half-cell with the high-potential ionogel electrolyte. d) First-cycle charge voltage profile of an NMC half-cell with the low-potential ionogel electrolyte. The voltage range and charge–discharge rate for the NMC half-cells were 2.5–4.3 V (vs Li/Li^+) and 0.1C, respectively.

low-potential ionogel electrolytes, which were obtained between −0.2 and 3.0 V (vs Li/Li^+) to evaluate their cathodic stability. The cathodic peak starting at 1.5 V (vs Li/Li^+) in the 1st cycle, which was observed for both the high-potential and low-potential ionogel electrolytes, can likely be attributed to impurities in the ionic liquids.[28] After the 1st cycle, the high-potential ionogel electrolyte showed a cathodic peak beginning at 0.7 V (vs Li/Li^+), which can be explained by reduction of the EMIM cations,[25] with the current density of this peak increasing with additional cycling, implying the instability of the high-potential ionogel electrolyte at low potentials. In contrast, the low-potential ionogel electrolyte showed suppression of the EMIM decomposition peak and stable current density during cycling since FSI anions enhance the reductive stability of EMIM cations.[29] Furthermore, the low-potential ionogel electrolyte showed a cathodic peak at <0 V (vs Li/Li^+) and an anodic peak at ≈0.2 V (vs Li/Li^+), stemming from the deposition and dissolution of lithium on the stainless-steel electrode (Figure S5, Supporting Information), respectively. On the other hand, the high-potential ionogel electrolyte exhibited an analogous cathodic peak at <0 V (vs Li/Li^+), but not a noticeable corresponding anodic peak, which suggests depletion of deposited lithium by reactions with the ionogel electrolyte because of its poor electrochemical stability at low potentials.

The low-potential stability of the ionogel electrolytes was also examined by lithium plating-stripping tests with Li|ionogel electrolyte|Cu cells, in which lithium was plated onto the copper electrode at a fixed current density and then stripped from the copper electrode at the same current density, as shown in Figure S6 in the Supporting Information. Because the redox reactions occur at ≈0 V (vs Li/Li^+) in this cell geometry, the tests allow the investigation of the ionogel electrolyte stability at low potentials by monitoring how much of the deposited lithium is recovered from the copper electrode. During the tests, low Coulombic efficiency (stripped lithium/plated lithium <18%) was observed with the high-potential ionogel electrolyte, indicating intense reactions between the deposited lithium and ionogel electrolyte at low potentials. Alternatively, high Coulombic efficiency (as high as 98%) was accomplished with the low-potential ionogel electrolyte, which further verifies the cathodic stability of this ionogel electrolyte. Moreover, to investigate the compatibility with low-potential anode materials, graphite half-cells were tested using the ionogel electrolytes, with a voltage range of 0.01–2.0 V (vs Li/Li^+) at 0.1C. As shown in Figure 3b, the charge cutoff voltage is lower than the cathodic limit of the high-potential ionogel electrolyte, precluding stable graphite intercalation chemistry at low potentials. However, the charge cutoff voltage is higher than the cathodic limit of the low-potential ionogel electrolyte, implying electrochemical compatibility

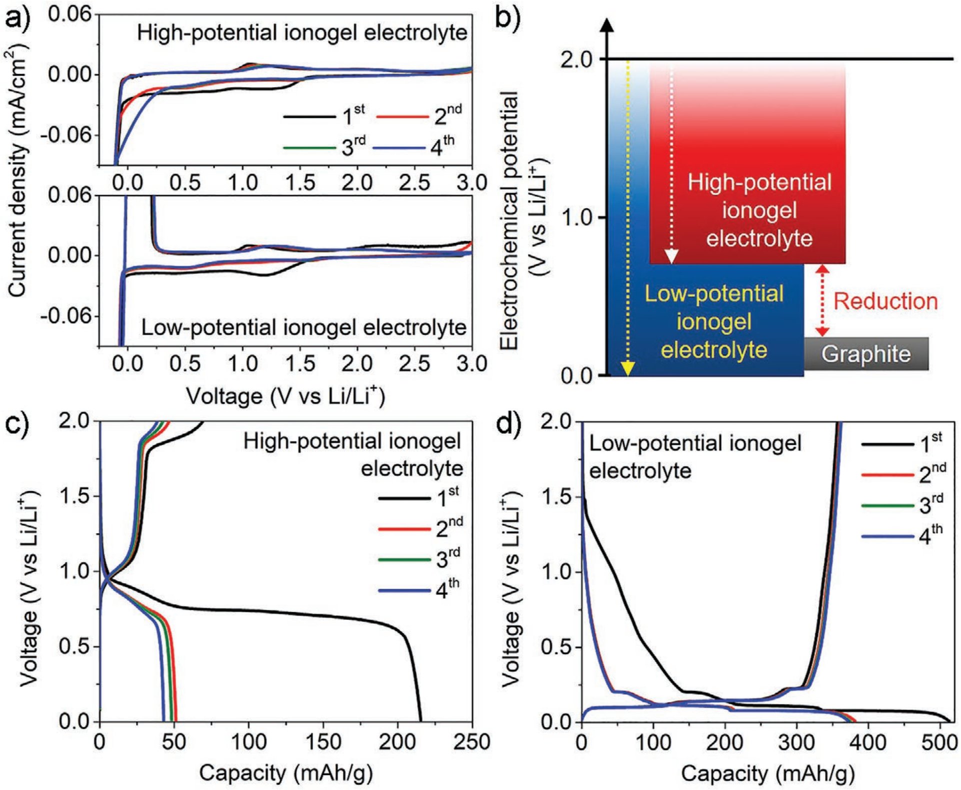

Figure 3. Cathodic stability of the high-potential and low-potential ionogel electrolytes. a) Cyclic voltammograms of Li|ionogel electrolyte|stainless-steel cells with (top) the high-potential and (bottom) low-potential ionogel electrolytes between –0.2 and 3.0 V (vs Li/Li^+) at a scan rate of 1 mV s^{-1}. b) Schematic diagram of the cathodic stability of the high-potential and low-potential ionogel electrolytes for graphite half-cells. c,d) Charge–discharge voltage profiles of graphite half-cells with the high-potential (c) and low-potential (d) ionogel electrolytes. The voltage range and charge–discharge rate for the graphite half-cells were 0.01–2.0 V (vs Li/Li^+) and 0.1C, respectively.

with the graphite anode. Indeed, while the graphite half-cell (Figure 3c) with the high-potential ionogel electrolyte exhibited low capacity and undesirable voltage profiles, the graphite half-cell (Figure 3d) with the low-potential ionogel electrolyte showed standard voltage profiles with well-defined plateaus (Figure S7, Supporting Information) and an initial discharge capacity of 357 mAh g^{-1}.

The CV and half-cell results suggest that the individual electrochemical windows of the high-potential and low-potential ionogel electrolytes do not concurrently cover the NMC and graphite potentials, suggesting that neither of these ionogel electrolytes alone are appropriate for NMC-graphite full-cells. Indeed, an NMC-graphite full-cell (Figure S8, Supporting Information) with the high-potential ionogel electrolyte exhibited insignificant discharge capacity and Coulombic efficiency because the poor cathodic stability of this ionogel electrolyte hinders the graphite anode from hosting lithium ions. Similarly, the NMC-graphite full-cell (Figure S9, Supporting Information) with the low-potential ionogel electrolyte revealed acute degradation during charging and was not fully charged because this ionogel electrolyte undergoes serious oxidation at the interface with the NMC cathode due to its low anodic stability.

To overcome these issues, NMC-graphite full-cells were fabricated using the layered heterostructure ionogel electrolyte, as depicted in **Figure 4**a. In this battery architecture (Figure S10, Supporting Information), the high-potential ionogel electrolyte contacts NMC to stabilize the interface with the cathode, and the low-potential ionogel electrolyte contacts graphite to stabilize the interface with the anode. In this manner, the layered heterostructure ionogel electrolyte offers an extended electrochemical window that fully covers the NMC cathode and graphite anode potentials in the full-cell geometry. Figure 4b displays the specific discharge capacity of this full-cell measured over multiple charge–discharge rates with a voltage range of 2.5–4.2 V. Regardless of the rate, the NMC-graphite full-cell exhibited typical voltage profiles (Figure 4c) and differential capacity curves (Figure 4d). In particular, the differential capacity curves showed two major peaks on charging, associated with lithium intercalation into graphite ($C_6 \rightarrow Li_xC_6$) and the NMC phase transition from a hexagonal to monoclinic (H1 → M) lattice[30,31] as well as their inverse peaks on discharging. Consequently, the differential capacity curves reveal that the layered heterostructure ionogel electrolyte induces desirable NMC-graphite full-cell chemistry. Furthermore, the initial discharge capacity of the full-cell (Figure 4b) was measured to be 123 mAh g^{-1} at 0.1C, and over 91% of the discharge capacity (Figure S11, Supporting Information) at 0.1C was retained at 1C, presenting favorable rate capability for a solid-state lithium-ion battery.

The cycling performance of the NMC-graphite full-cell based on the layered heterostructure ionogel electrolyte was evaluated at 0.5C, as shown in **Figure 5**a. A control cell was also tested using an ionogel electrolyte prepared with mixed ionic liquids (i.e., 50% EMIM-TFSI/LiTFSI and 50% EMIM-FSI/LiTFSI by

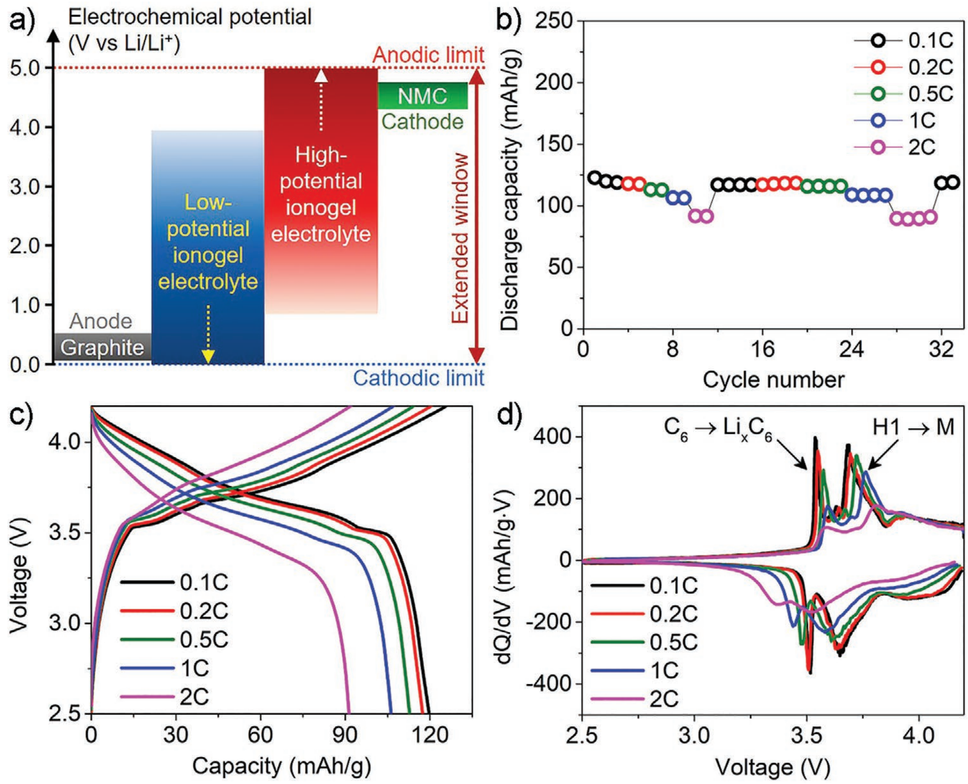

Figure 4. NMC-graphite full-cells based on layered heterostructure ionogel electrolytes. a) Schematic diagram of the electrochemical window of the layered heterostructure ionogel electrolyte for NMC-graphite full-cells. b) Specific discharge capacity of the NMC-graphite full-cell at different rates, with a voltage range of 2.5–4.2 V. Charge rates are the same as the discharge rates. c) Charge–discharge voltage profiles and d) corresponding differential capacity (dQ/dV) curves for the NMC-graphite full-cell at different rates.

weight) to elucidate the importance of layering the ionic liquids for the expanded electrochemical window. Compared to the full-cell with the layered heterostructure ionogel electrolyte, the control cell showed comparable initial capacity, but the capacity faded significantly with lower Coulombic efficiency and considerable voltage profile degradation (Figure 5b and Figure S12, Supporting Information). Similar results were also observed for an NMC-graphite full-cell (Figure S13, Supporting Information) using the mixed ionic liquid electrolyte without hBN nanoplatelets, revealing that the influence of the hBN matrix on the electrochemical performance was limited. The poor performance of the control cell can be attributed to the mixed electrolyte structure, in which both the high-potential and low-potential ionic liquids are in contact with the cathode and anode (Figure 5c). The mixed structure leads to side reactions between the high-potential (or low-potential) ionic liquid and anode (or cathode), which results in rapid cell degradation. However, the full-cell with the layered heterostructure ionogel electrolyte exhibited superior cycling performance (Figure 5a and Figure S14, Supporting Information) with minimal changes in the voltage profiles (Figure 5d) during cycling since the layered structure (Figure 5e) prevents each ionic liquid from contacting its incompatible electrode. Furthermore, the strong interactions between the ionic liquids and the hBN nanoplatelets ensure that they remain localized in their respective layers, thus minimizing intermixing even following extended cycling. Therefore, the cycling performance reveals that the layered heterostructure ionogel electrolyte enables the extension of electrochemical windows for solid-state lithium-ion batteries, while minimizing the side effects of combining two ionic liquids.

3. Conclusion

We have developed layered heterostructure ionogel electrolytes with two different imidazolium ionic liquids and hBN nanoplatelets to achieve wide electrochemical stability for solid-state lithium-ion batteries. In this approach, each ionogel electrolyte provides a different electrochemical window to allow stability against both the high-potential cathode and the low-potential anode, with the hBN nanoplatelets providing a large surface area to immobilize the ionic liquids and thus minimize intermixing at the heterointerface. The combined electrochemical windows enable the fabrication of full-cell solid-state lithium-ion batteries with voltages that are unachievable with the individual ionic liquids. Compared to ionogel electrolytes using mixed ionic liquids, the layered heterostructure ionogel electrolyte allows the extension of electrochemical windows without the side effects of combining two different ionic liquids, resulting in significantly enhanced cycling performance. Ongoing efforts will explore the ion exchange kinetics at the heterointerfaces of the layered heterostructure ionogel electrolytes in an effort to determine the limits of this methodology. Overall, while demonstrated here for solid-state lithium-ion batteries, this layered heterostructure ionogel electrolyte approach can likely be generalized to other emerging solid-state battery technologies.[32]

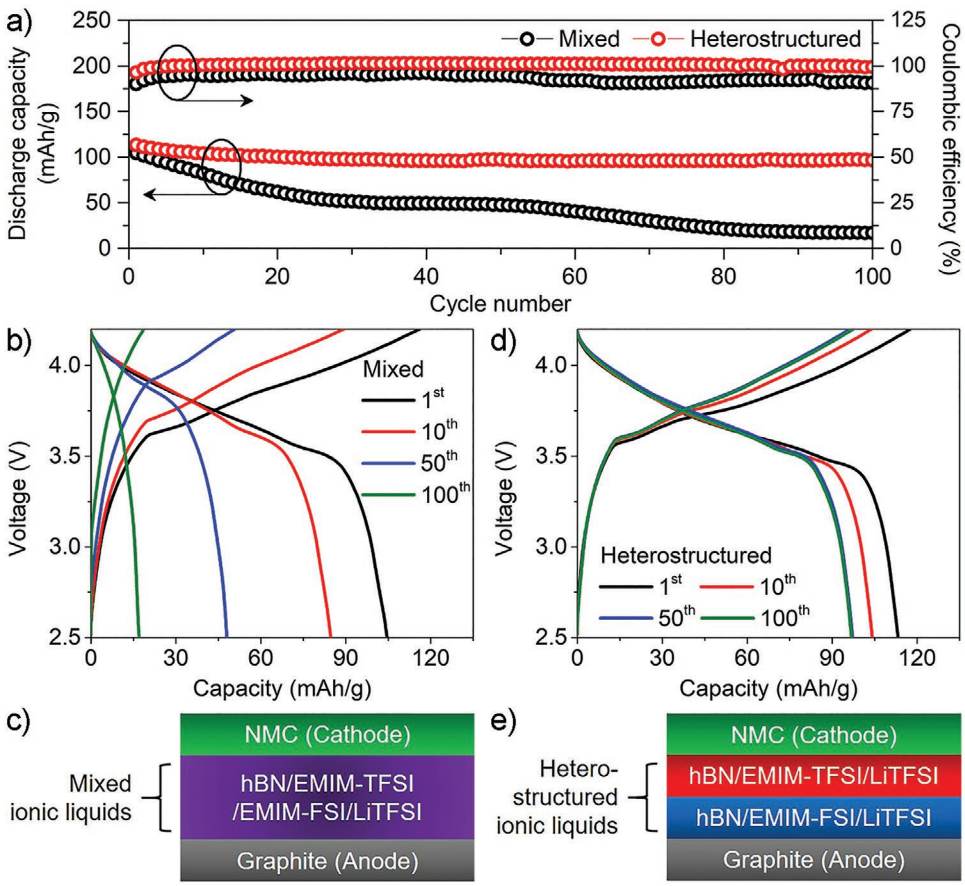

Figure 5. Comparison of NMC-graphite full-cells with mixed and layered heterostructure ionogel electrolytes. a) Discharge capacity and Coulombic efficiency of NMC-graphite full-cells with mixed and layered heterostructure ionogel electrolytes at a charge-discharge rate of 0.5C. The mixed ionogel electrolyte was prepared with 50% EMIM-TFSI/LiTFSI and 50% EMIM-FSI/LiTFSI by weight. b) Charge–discharge voltage profiles and c) schematic of the NMC-graphite full-cell with the mixed ionogel electrolyte. d) Charge–discharge voltage profiles and e) schematic of the NMC-graphite full-cell with the layered heterostructure ionogel electrolyte.

4. Experimental Section

Exfoliation of hBN Nanoplatelets: hBN nanoplatelets were prepared from bulk hBN microparticles by a liquid-phase exfoliation method.[27] In a typical batch process, 120 g of bulk hBN microparticles (≈1 µm, Sigma-Aldrich), and 12 g of ethyl cellulose (4 cP viscosity grade, Sigma-Aldrich) were added to 800 mL of ethanol, where ethyl cellulose was used as a dispersing agent to improve the exfoliation yield.[33] The solution was shear-mixed at 10,230 rpm for 2 h, using a rotor/stator mixer (L5M-A, Silverson) with a square hole screen. The shear-mixed solution was centrifuged (J26-XPI, Beckman Coulter) at 4000 rpm for 20 min to sediment large particles. The supernatant was collected and mixed with an aqueous solution of 40 mg mL^{-1} sodium chloride (16:9 by weight) to flocculate exfoliated hBN nanoplatelets and ethyl cellulose, followed by centrifugation at 7500 rpm for 6 min. The sedimented hBN nanoplatelets and ethyl cellulose were washed with deionized water to eliminate residual sodium chloride, dried with an infrared lamp, and ground with a mortar and pestle to yield a powder. Finally, the powder was annealed at 400 °C for 4 h in air to decompose ethyl cellulose, leaving behind a thin oxidized amorphous carbon coating on the surface of the hBN nanoplatelets that induces stronger solidification of ionogels.[27]

Formulation of Ionogel Electrolytes: 1 M LiTFSI (99.95% trace metal basis, Sigma-Aldrich) was dissolved in EMIM-TFSI (H_2O ≤ 500 ppm, Sigma-Aldrich) and EMIM-FSI (H_2O ≤ 0.002%, Solvionic) by stirring with a magnetic stir bar on a hotplate at 60 °C for 24 h. Employing a mortar and pestle, the EMIM-TFSI/LiTFSI and EMIM-FSI/LiTFSI were mixed with the hBN nanoplatelets to formulate high-potential and low-potential ionogel electrolytes, respectively. After mixing, the ionogel electrolytes (hBN and ionic liquids) formed highly viscous materials, as shown in Figures S1 and S2 in the Supporting Information, which were then used for characterization and battery assembly. To prepare the mixed ionogel electrolyte, the ionic liquids (50% EMIM-TFSI/LiTFSI and 50% EMIM-FSI/LiTFSI by weight) and hBN nanoplatelets were mixed using the mortar and pestle. The ratio between the ionic liquids and hBN nanoplatelets for all ionogel electrolytes was 3:2 by weight.

Characterization: Exfoliated hBN nanoplatelets were observed using a scanning electron microscope (SU8030, Hitachi). The ionic conductivity (σ) of the ionogel electrolytes was measured with a stainless-steel|ionogel electrolyte|stainless-steel geometry and the following equation

$$\sigma = \frac{t}{R \times A} \tag{1}$$

where t and A are the thickness and area, respectively, of the ionogel electrolyte between the stainless-steel electrodes. In addition, R is the bulk resistance determined by electrochemical impedance spectroscopy using a potentiostat (VSP, BioLogic), with a frequency range of 1 MHz–100 mHz and an amplitude of 10 mV.[34] To investigate the lithium-ion transference number (T_{Li}), lithium symmetric cells were assembled with the ionogel electrolytes and polarized at 10 mV. Their Nyquist plots were acquired before and after polarization and T_{Li} was calculated based on the following equation[35]

$$T_{Li} = \frac{I_S(\Delta V - I_0 R_0)}{I_0(\Delta V - I_S R_S)} \tag{2}$$

where ΔV, I, and R are the applied polarization voltage, current from polarization curves, and charge-transfer resistance from the Nyquist plots. Subscripts 0 and S denote the initial and steady states, respectively. The electrochemical windows of the high-potential and low-potential ionogel electrolytes were characterized with Li|ionogel electrolyte|stainless-steel coin cells (CR2032-type) by cyclic and linear sweep voltammetry using the potentiostat at a scan rate of 1 mV s^{-1} at room temperature. Different cells were used to investigate the anodic and cathode limits. Lithium plating-stripping tests were performed using Li|ionogel electrolyte|Cu coin cells (CR2032-type) after cycling the cells between 0 and 0.5 V (vs Li/Li^+) for ten cycles.[36,37] Viscoelastic properties of the ionogel electrolytes were characterized using a rheometer (MCR 302, Anton Paar) equipped with a 8 mm diameter parallel plate (gap between the rheometer stage and parallel plate: 1 mm) with a strain of 0.1%. The viscosity of the ionic liquids was measured using the rheometer equipped with a 25 mm, 2° cone, and plate geometry at 25 °C.

Preparation of Electrodes: $LiNi_{0.5}Mn_{1.5}O_4$, $LiNi_{0.33}Mn_{0.33}Co_{0.33}O_2$, and graphite (natural, 325 mesh, 99.8% metal basis) were purchased from MTI Corporation, Targray, and Alfa Aesar, respectively. The active materials, carbon black (MTI Corporation), and poly(vinylidene fluoride) (MTI Corporation or Kynar) in a weight ratio of 8:1:1 were mixed with 1-methyl-2-pyrrolidinone. The slurries were cast onto aluminum (for $LiNi_{0.5}Mn_{1.5}O_4$ and $LiNi_{0.33}Mn_{0.33}Co_{0.33}O_2$) or copper (for graphite) substrates, followed by drying in a vacuum oven at 80 °C for longer than 24 h. The electrodes were used after cutting into circles with a diameter of 1 cm. Active material loading was 4.0, 2.1, and 0.9 mg cm^{-2} for the $LiNi_{0.5}Mn_{1.5}O_4$, $LiNi_{0.33}Mn_{0.33}Co_{0.33}O_2$, and graphite electrodes, respectively.

Battery Assembly: CR2032-type coin cells were used for all battery testing. To enhance the interfacial contact between the ionogel electrolytes and electrodes, a small amount (≈10 mg) of ionic liquid was drop-cast onto the electrodes and the excess on the electrode surfaces was removed with a Kimtech wipe. For half-cells, ionogel electrolytes were transferred and spread on lithium metal using a spatula, and NMC and graphite electrodes were placed on the ionogel electrolytes. NMC and graphite electrodes for full-cells were pretreated to minimize the irreversible capacity of initial cycles. For the pretreatment, half-cells were assembled with a glass microfiber filter (GF/C grade, Whatman) and the ionic liquid electrolyte (EMIM-TFSI/LiTFSI for NMC and EMIM-FSI/LiTFSI for graphite), charged and discharged for one cycle at 0.05C, and disassembled to recover the electrodes. For full-cells with the layered heterostructure ionogel electrolyte (Figure S10a, Supporting Information), the high-potential and low-potential ionogel electrolytes were applied to NMC and graphite electrodes, respectively, and the electrodes were sandwiched. The same mass (≈20 mg) of ionogels was used for both the high-potential and low-potential electrolytes to minimize thickness variation. For control cells, the mixed ionogel electrolyte was deposited onto NMC electrodes and graphite electrodes were placed on the ionogel electrolyte. The thicknesses of the layered heterostructure and mixed ionogel electrolytes were 60–80 μm, which were measured after disassembling the full-cells. However, it is expected that the electrolyte thickness can be considerably reduced by depositing the ionogels with printing methods.[38] All full-cells were tested after one activation cycle at 0.5C. All battery cells were prepared in an argon-filled glovebox and measured with a battery testing system (BT-2143, Arbin) at room temperature.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

electrochemical stability window, ion gels, ionic liquids, lithium-ion batteries, solid-state electrolytes

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